

PLAIN LANGUAGE GUIDE

Pharmacovigilance Made Simple.

The story of Martin, a medicine called Relvian, and how the global drug safety system works.

9 chapters · Plain language · Free download



What is pharmacovigilance?

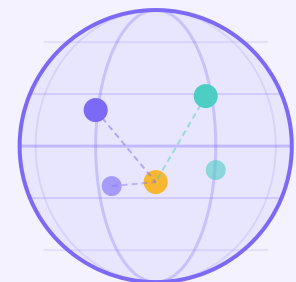
REGULATORY DEFINITION

"The science and activities relating to the detection, assessment, understanding and prevention of adverse effects or any other medicine-related problem."

Article 1(28d), Directive 2001/83/EC as amended by Directive 2010/84/EU

Once a medicine is authorised and reaches patients, scientific scrutiny does not cease. Real-world patient populations differ substantially from clinical trial cohorts: they are older, carry multiple comorbidities, and are frequently prescribed concurrent therapies. Certain adverse effects are too infrequent to manifest during pre-authorisation studies; others emerge only after prolonged exposure or in specific patient subgroups.

Pharmacovigilance is the regulatory and scientific framework that identifies, evaluates, and responds to these risks throughout a medicine's entire authorised lifecycle. It depends on healthcare professionals, patients, manufacturers, and regulatory authorities each fulfilling a defined and legally mandated role.



VIGIBASE

Global Safety Database
WHO, Uppsala

130+

countries in the WHO
International Drug Monitoring
Programme

30M+

individual case safety reports in
VigiBase

This is Martin.

Martin is 61 years old and has osteoarthritis in his knees. His pain has gradually worsened, and it is no longer well controlled with exercise and occasional over-the-counter pain relief.

REGULATORY CONTEXT

"Patient" means any individual who uses or receives healthcare. Patients are considered key stakeholders in pharmacovigilance reporting under GVP Module VI (Rev 2).

EMA GVP Module VI Rev 2, 2017

HIS DIAGNOSIS

Osteoarthritis is a degenerative joint disease characterised by cartilage breakdown, pain, and reduced mobility. It affects more than 500 million people worldwide and is among the leading causes of chronic pain in adults over 50.



Medical History and Treatment Initiation.

Martin's doctor prescribes him Relvian, a new anti-inflammatory medicine that has just come on the market. Martin collects the medicine from his local pharmacy. The pharmacist informs him of the known potential side effects: indigestion, dizziness, and mild headache.

After a week, Martin develops unusual shortness of breath when climbing stairs and a persistent dry cough. He also feels more tired than usual. He wonders if it has something to do with his new medicine. He makes an appointment with his doctor to find out.

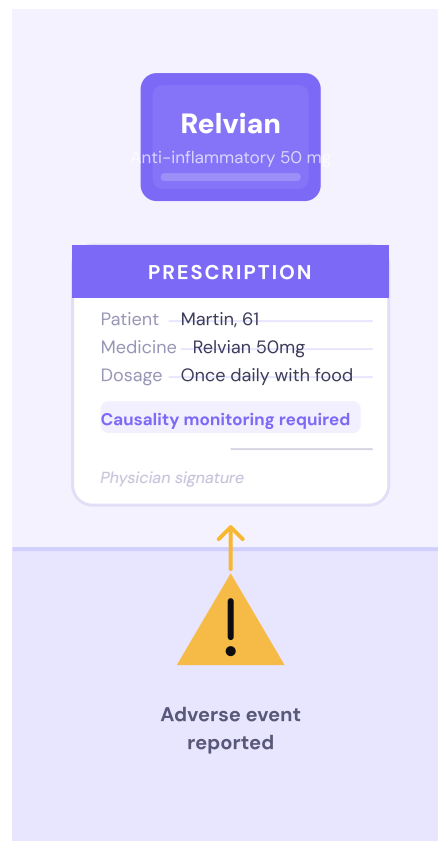
REGULATORY CONTEXT

"Member States shall take all appropriate measures to encourage patients, doctors, pharmacists and other healthcare professionals to report suspected adverse reactions to the competent authority."

Article 107a, Directive 2001/83/EC

KNOWN SIDE EFFECTS AT THIS POINT

Indigestion, dizziness, and mild headache. Drug-induced lung inflammation was not listed as a known adverse effect of Relvian at the time of prescription.



Clinical Evaluation.

Martin's doctor examines him and orders tests, including chest imaging and blood work. The results show signs of drug-induced lung inflammation (pneumonitis). He checks that Martin is not taking any other medications that could be the cause. He also orders more tests to rule out infection or underlying lung disease. He advises Martin to stop taking Relvian.

Over the next two weeks, Martin's breathing gradually improves, and the cough disappears. His oxygen levels return to normal.

REGULATORY CONTEXT

"Serious adverse reaction means an adverse reaction which results in death, is life-threatening, requires inpatient hospitalisation or prolongation of existing hospitalisation, results in persistent or significant disability or incapacity..."

Article 1(12), Directive 2001/83/EC

DRUG-INDUCED PNEUMONITIS

Inflammation of lung tissue caused by a medicine. In Martin's case, the temporal association, the resolution of symptoms after discontinuation, and the absence of alternative explanations pointed to Relvian as the probable cause.



Individual Case Safety Report (ICSR) Submission.

Martin's doctor fills out a report for the adverse event Martin has experienced, as he suspects Relvian may be the cause. Drug-induced pneumonitis was not listed as a known side effect. He sends the report to the national regulatory authority.

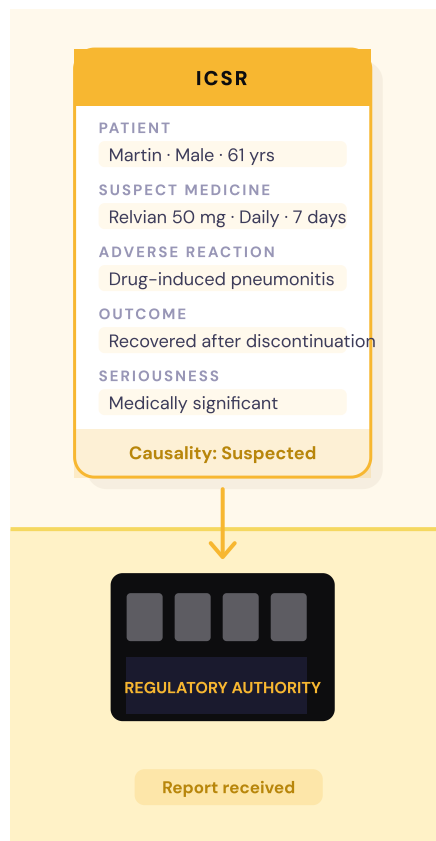
REGULATORY DEFINITION

"Individual case safety report (ICSR) means a format for the reporting of one or more suspected adverse reactions to a medicinal product that occur in a single patient at a specific point of time."

Article 1(16a), Directive 2001/83/EC

WHAT THE ICSR CAPTURES

Patient demographics, the suspect medicine with dose and duration, a detailed description of the adverse reaction, the outcome, and the reporter's causality assessment. This structured format enables global databases to analyse and compare cases across countries.



Safety Database Entry and Signal Detection.

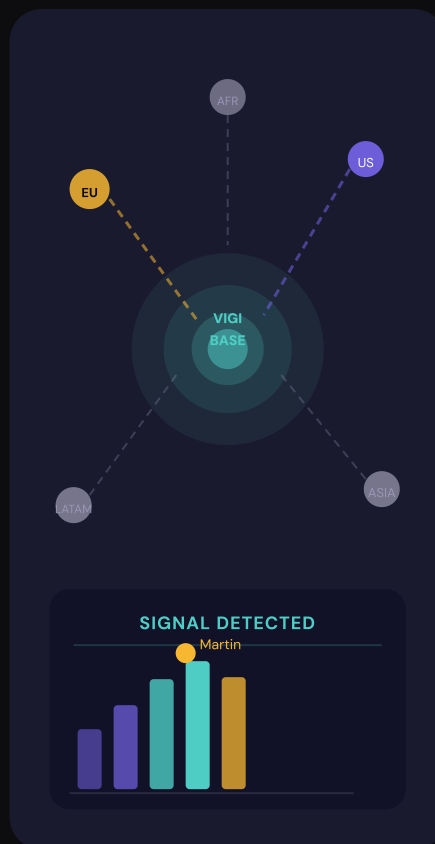
REGULATORY DEFINITION

"Signal means information arising from one or multiple sources, including observations and experiments, which suggests a new potentially causal association, or a new aspect of a known association, between an intervention and an event or set of related events..."

GVP Module IX Rev 1, EMA, 2017

Martin's case is added to the national database of adverse event reports for medicines and vaccines. Since the regulatory authority in Martin's country is a member of the WHO Programme for International Drug Monitoring, his report is shared with the global database, VigiBase.

At the regulatory authority, Martin's report is reviewed by a case assessor who finds a small number of similar reports involving Relvian and lung inflammation in both the national database and VigiBase. Further analysis suggests it is probable that Relvian may be associated with drug-induced pneumonitis.



Regulatory Action, Label Update and Risk Communication.

The regulatory authority decides that Relvian's manufacturer must update the product information. "Inflammation of the lungs" is added to the list of adverse effects, and a warning is included about the potential risk of serious respiratory reactions.

An advisory letter is sent to health providers, with additional information made available on the regulatory authority's website. Martin's doctor is informed and reviews prescribing guidance.

REGULATORY DEFINITION

"Risk minimisation measures are public health interventions intended to prevent or reduce the occurrence of adverse reactions associated with the exposure to a medicine..."

GVP Module XVI Rev 2, EMA, 2017

LABEL UPDATE

"Inflammation of the lungs" added to Relvian's Summary of Product Characteristics.

WARNING INCLUDED

Risk of serious respiratory reactions added to the special warnings and precautions section.

ADVISORY LETTER SENT

Direct healthcare professional communication issued.
Information published on the regulatory authority's website.

BEFORE UPDATE

Adverse effects:

- Indigestion
- Dizziness
- Mild headache

Lung inflammation: not listed

Respiratory warning: absent

LABEL UPDATED

AFTER UPDATE

Adverse effects:

- Indigestion
- Dizziness
- Mild headache

· Lung inflammation — NEW

Warning: serious respiratory risk added

Advisory letter issued to prescribers

Pharmacovigilance Outcome and Risk Minimisation.

Martin's case contributes to a better understanding of Relvian's safety profile. Reports like this help identify rare risks and improve safe use of new medicines.

Martin's doctor switches him to a different osteoarthritis treatment — pain management with physical therapy support and an alternative analgesic — and Martin remains stable with no further side effects.

"Martin did not do anything extraordinary. His doctor did not do anything extraordinary. The pharmacovigilance system worked exactly as it was designed to."

WHY THIS MATTERS

Clinical trials are conducted in controlled conditions with selected patient populations. Post-authorisation pharmacovigilance identifies signals that cannot be detected before approval. Reports like Martin's directly fulfil the legal obligation to protect public health throughout a medicine's authorised lifecycle.



Glossary.

All terms from Martin's story, defined with their regulatory source.

PHARMACOVIGILANCE

The science and activities relating to the detection, assessment, understanding and prevention of adverse effects or any other medicine-related problem. (Art. 1(28d), Dir. 2001/83/EC)

ADVERSE REACTION

A response to a medicinal product which is noxious and unintended. Implies a causal relationship. Distinct from an "adverse event," which does not imply causality. (Art. 1(11), Dir. 2001/83/EC)

ICSR

A format for reporting one or more suspected adverse reactions to a medicinal product that occur in a single patient at a specific point in time. (Art. 1(16a), Dir. 2001/83/EC)

SIGNAL

Information suggesting a new potentially causal association between an intervention and an event, or a new aspect of a known association, warranting further investigation. (GVP Module IX Rev 1)

VIGIBASE

The WHO global database of individual case safety reports, managed by the Uppsala Monitoring Centre. Contains over 30 million reports from more than 130 member countries.

RISK MINIMISATION

Public health interventions intended to prevent or reduce the probability of occurrence of an adverse reaction, or to reduce its severity or impact. (GVP Module XVI Rev 2)

SUMMARY OF PRODUCT CHARACTERISTICS

The official document approved by a regulatory authority that forms the basis of information for healthcare professionals on how to use a medicinal product safely and effectively.

DRUG-INDUCED PNEUMONITIS

Inflammation of lung parenchyma caused by a pharmaceutical agent. May present as cough, dyspnoea, and hypoxia. Diagnosis requires exclusion of infection and other aetiologies.



Martin's report made medicines safer for everyone.

He did not do anything extraordinary. His doctor did not do anything extraordinary. The pharmacovigilance system worked exactly as it was designed to. Every report matters. Every patient counts.

Qualivio supports pharmacovigilance professionals and organisations with education, consulting, and practical tools built around current regulatory frameworks.

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